

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni		Omni		

Section 3

Cabinet installation

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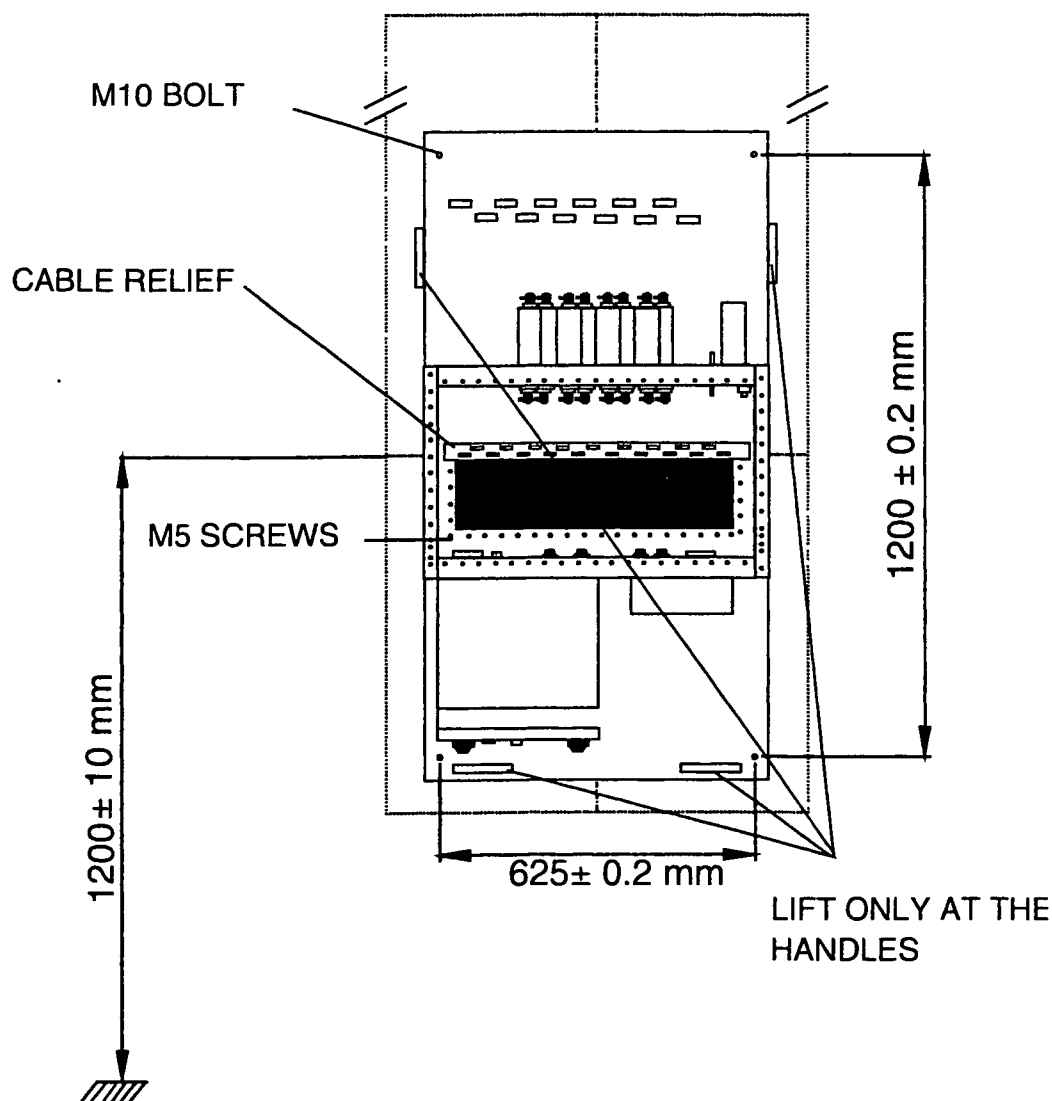
Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni		Omni		

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
Standard	Omni	Omni

1. MOUNTING THE SYSTEM FILTER BOX

1. Remove the dummy plate used for testing the RF shielded room. Clean the contact areas of the S.F.B. and of the RF enclosure.
2. Remove the upper aluminum covers from the System Filter Box to gain access to the handles. Also remove the middle copper cover.
3. Clean the contact area's of the RF cage interface and of the system filter box.
4. Hang the System Filter Box on the M10 bolts (do not tighten) prepared by the RF supplier. Only lift at the handles.
5. Mount all M5 screws with washers delivered by RF cage supplier.
6. Tighten the M10 nuts

Figure 1 - Mounting the System Filter Box

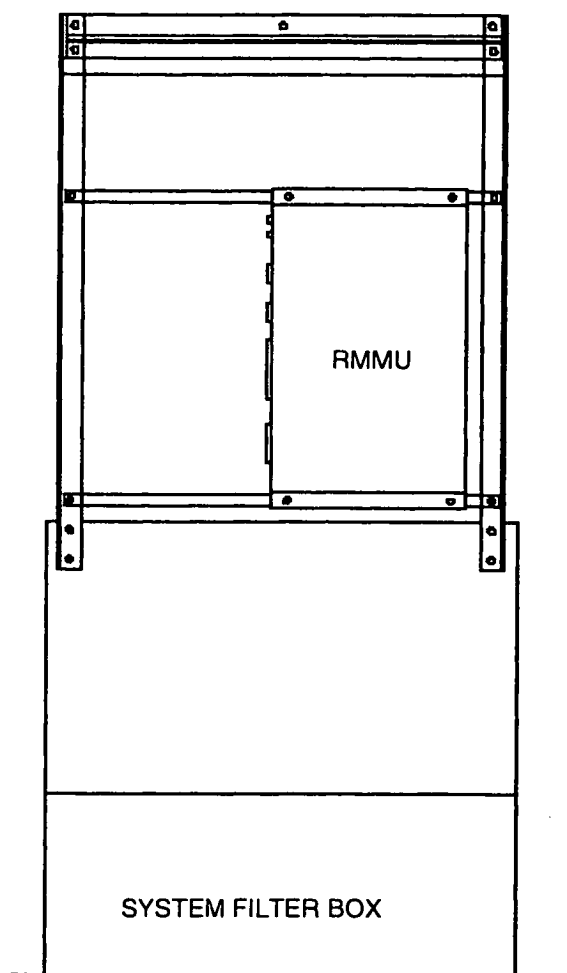


Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni		Omni		

2. MOUNTING THE REMOTE MAGNET MONITORING UNIT (RMMU)

The RMMU must be mounted on the topframe of the System Filter Box (SFB).

Figure 2: Position of RMMU



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Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
Standard	Omni	Omni

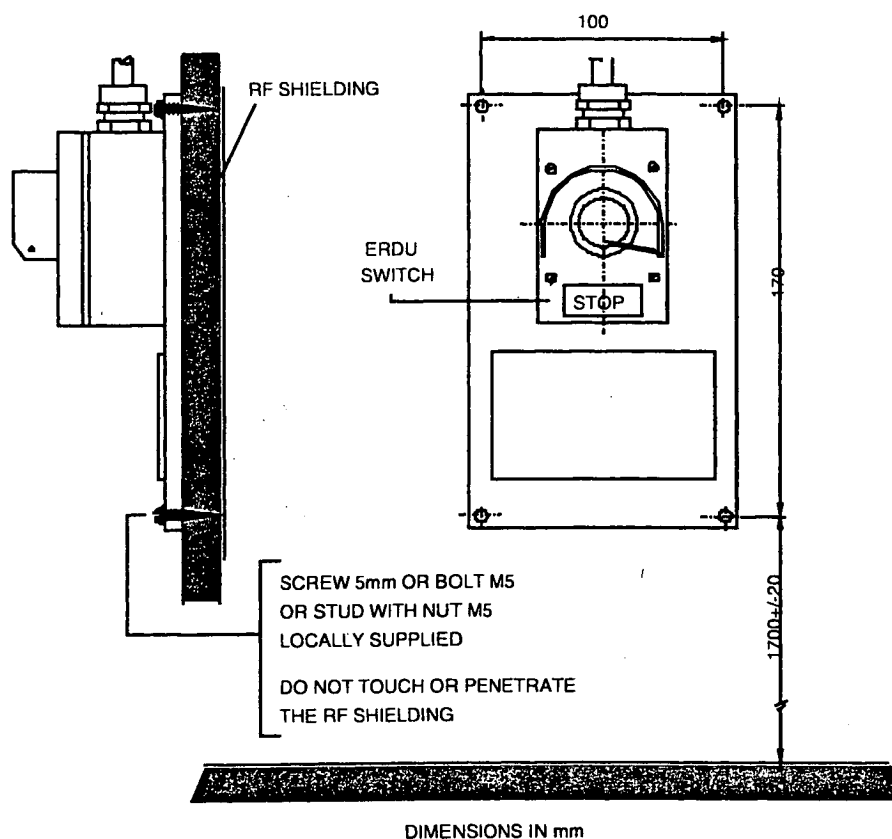
3. MOUNTING THE EMERGENCY RUN DOWN UNIT (ERDU) SWITCHES

One of the ERDU switches must be mounted inside the RF shielded room and one in the control room. Mount the ERDU switch inside the RF room to the planned position (see drawings of the planning department). Ask the RF room supplier how to mount the ERDU switch in order not to damage the RF shield.

CAUTION

Be sure not to penetrate or make contact with the RF shielding when installing the ERDU switch inside the RF room.

Figure 3: Mounting the ERDU switch

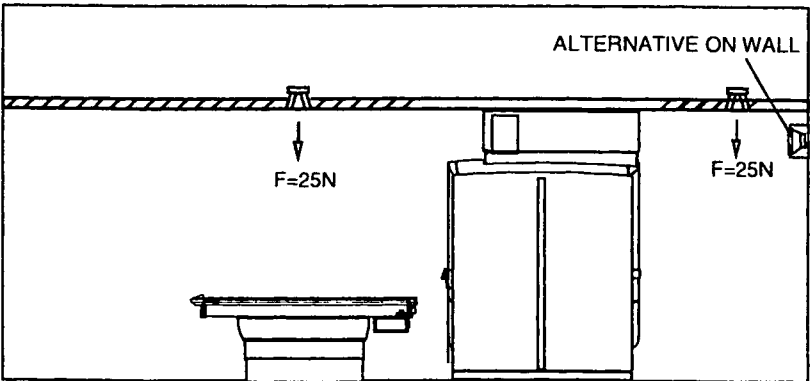


Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
Standard	Omni	Omni

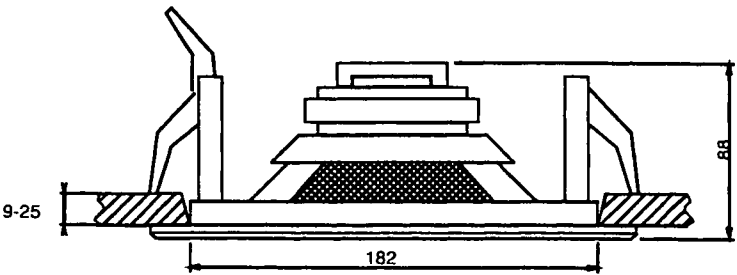
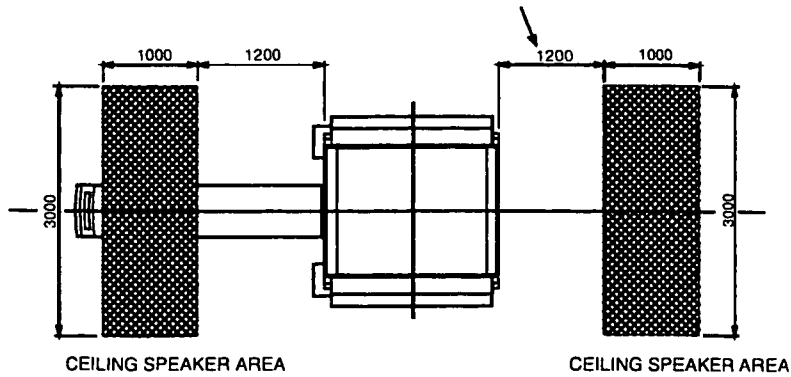
4. MOUNTING THE CEILING SPEAKERS

Two ceiling speakers have to be mounted. One above the patient support and the other one at the rear of the magnet (see figure). The distance of 1.20m is an average in order to obtain a good sound quality on both sides of the magnet. The minimum distance at the rear side is 0.3m. In that case, the quality is decreases but is still acceptable. As an alternative, it is possible to mount the speaker on the wall. Use one of the delivered build-on boxes.

Figure 4 - Mounting positions of the ceiling speakers



MINIMAL 550mm IN CASE OF MINIMAL ROOM LAYOUT



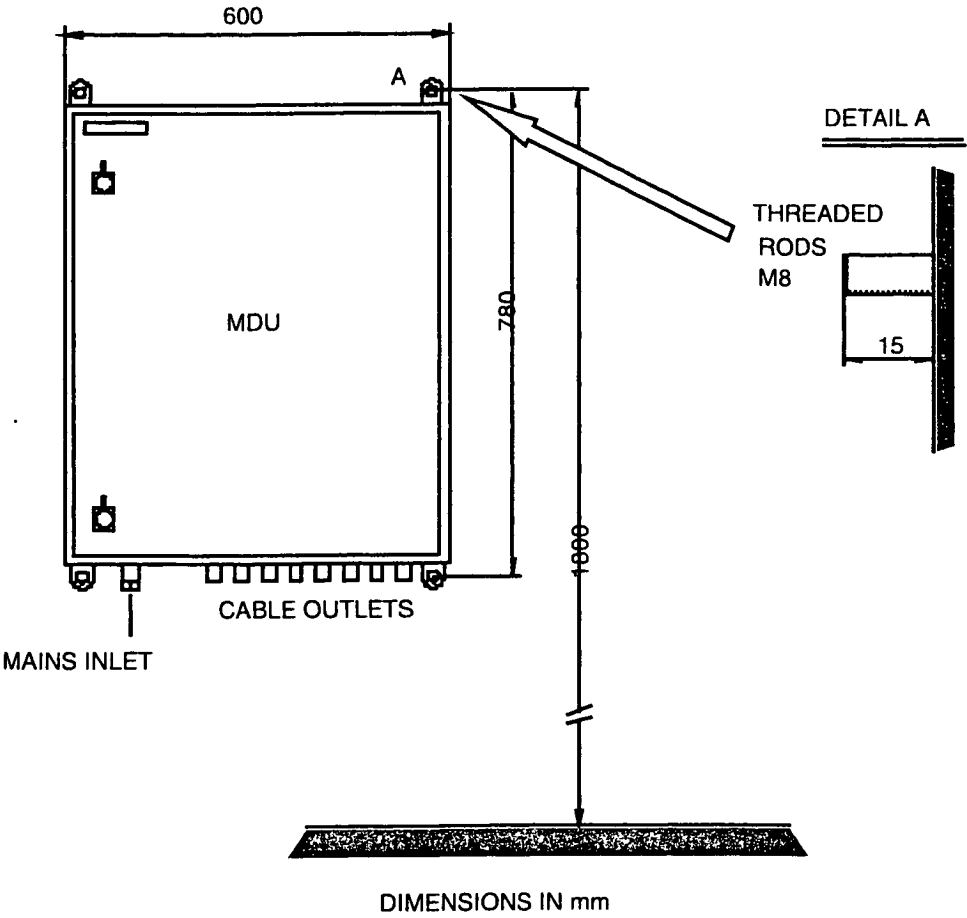
Dimensions in mm

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
Standard	Omni	Omni

5. MOUNTING THE MAINS DISTRIBUTION UNIT (MDU)

The MDU must be mounted to four threaded rods mounted in the technical room. Arrange a local electrician to connect the hospital mains to the MDU

Figure 5 - Mounting the MDU

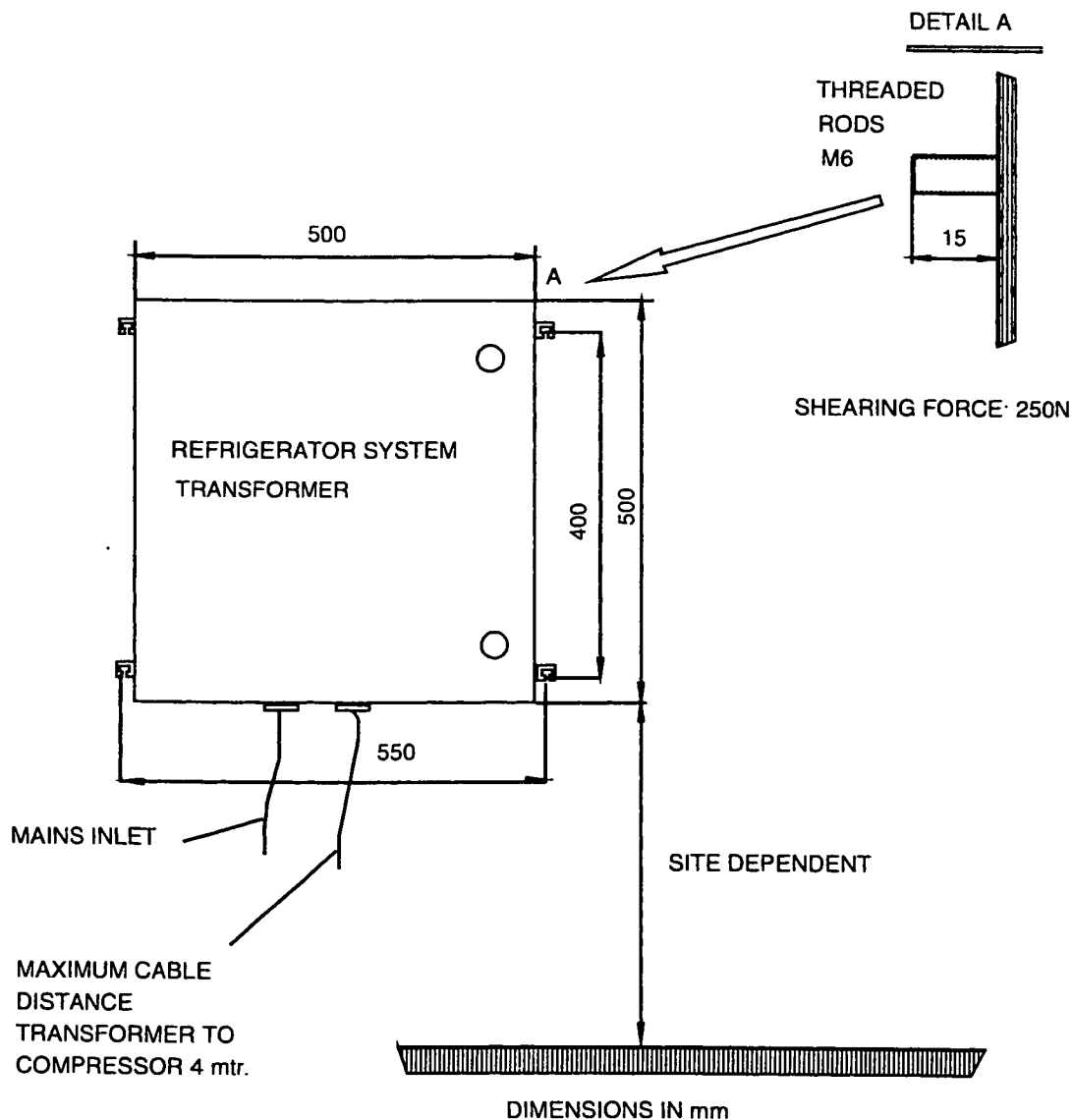


Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
Standard	Omni	Omni

6. MOUNTING THE REFRIGERATOR SYSTEM TRANSFORMER (NOT FOR USA)

The refrigerator system transformer must be mounted to four threaded rods in the technical room.

Figure 6: Mounting the refrigerator system transformer



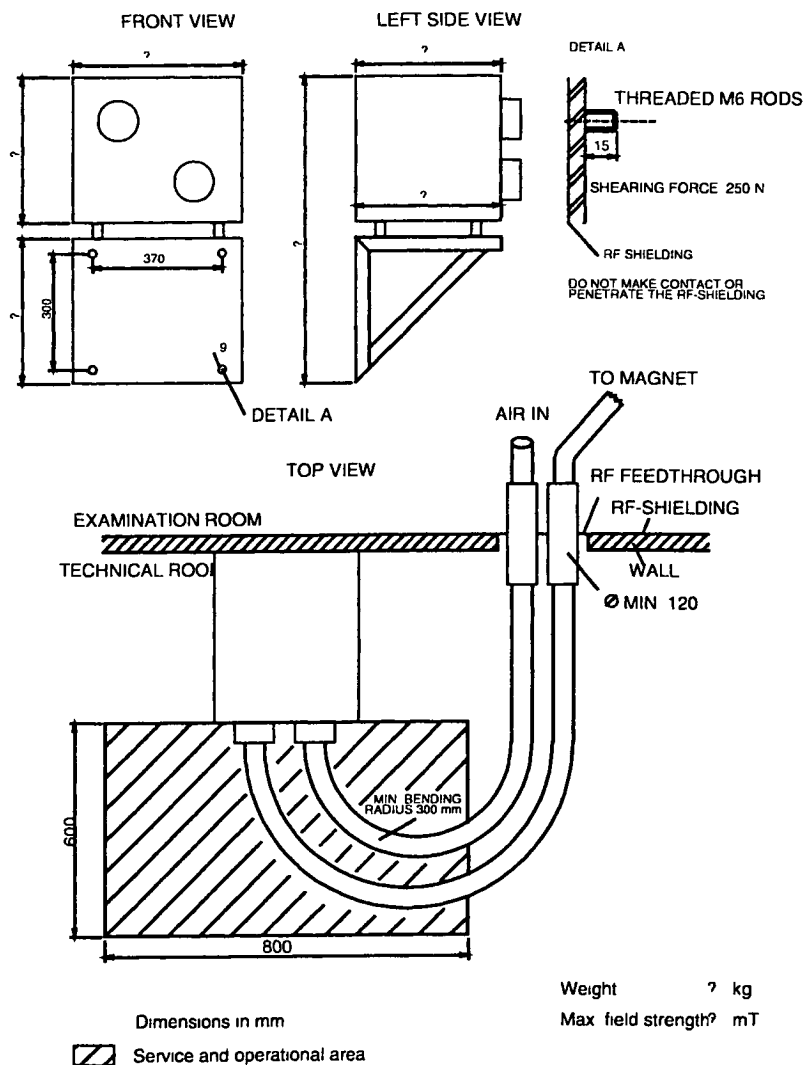
FH127 DRW

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
Standard	Omni	Omni

7. MOUNTING THE PATIENT COOLING UNIT

Mount the patient-cooling unit against the wall inside the technical room.
Two mounting brackets are supplied with the unit. Mount these brackets to the wall.
Mount the cooling unit on top of these brackets using four dampers.
An air filter must be installed at the air outlet of the patient-cooling unit.

Figure 7 – Mount patient cooling unit



Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
Standard	Omni	Omni

8. POSITIONING CABINETS

Position all cabinets into the technical room according to the Site-Planning drawings. For dimensions of cabinets and service area consult the Intera Planning Reference Book.

Level all cabinets using a spirit level and tighten all feet.

8.1. RF-AMPLIFIER

For the Intera 0.5T the MR5002 RF-amplifier is installed inside the NTDAC at the bottom of the cabinet.

For Intera 1.0T and 1.5T the RF amplifier must be positioned next to the NTDAC.

Release independent, a new (smaller) RF amplifier will be supplied.
This is the S23 for the Intera 1.5T and the S24 for the Intera 1.0T.

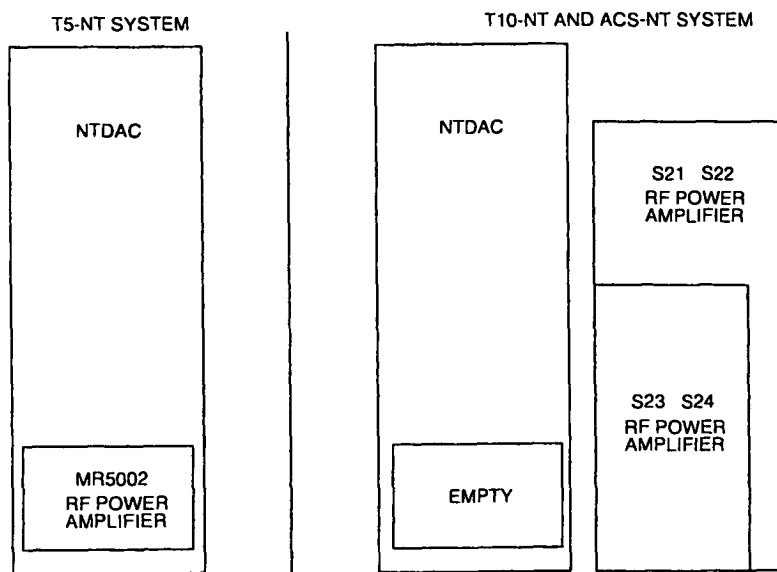
NOTE

*The Mains Power (three phase) for the S21, S22 and MR5002 RF amplifiers comes from the NT-DAC.
However, the Mains Power (single phase) for the S23 and S24 RF amplifiers comes directly from the MDU.*

NOTE

The RF-Output on the S23 and S24 RF amplifiers is located on the rear of the amplifier.

Figure 8: Location of RF Amplifier



FH128 DRW

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
Standard	Omni	Omni

8.2. LINE FREQUENCY (50/60 Hz) SELECTION (S21 AND S22 AMPLIFIERS ONLY).

The S21-64 and S22-42 RF amplifiers must be adjusted to 50 or 60 Hz.

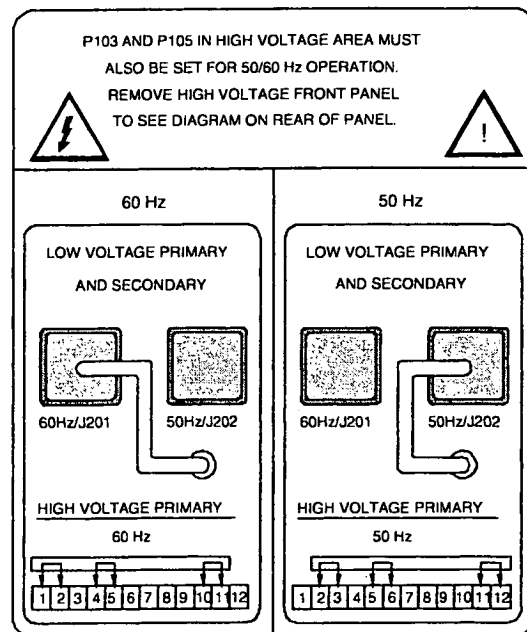
CAUTION

S21-64 and S22-42 RF Amplifier Line Frequency (50/60 Hz) Selection.

The S21/22 RF amplifier incorporates ferro-resonant supplies, which must be configured for either 50 Hz or 60 Hz operation. To re-configure the amplifier from one frequency to the other:

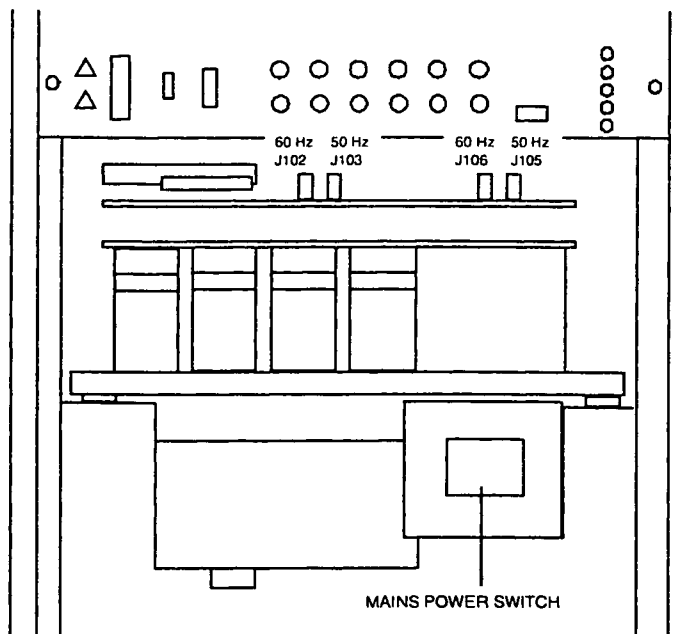
1. If not yet done, switch off the mains power to the RF amplifier. Disconnect the mains power plug of the RF amplifier from the NT-DAC.
2. Remove the 50/60 Hz service panel from the rear of the RF amplifier and follow the directions located on the panel. Replace the panel when complete.

Figure 9: S21/22 Rear Panel screening



3. Remove the panel covering the HV assembly. This is the lower front panel. Take care to unplug the power supply fan when removing the panel. Its connector is located behind the panel on the lower right hand side. Attach a shorting strap between PA HV+ and PA HV- on the HV Rectifier board. Also connect a shorting strap between IPA HV+ and IPA HV- on the HV Rectifier board.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
Standard	Omni	Omni

Figure 10: S21/22 HV Assembly

Locate P103 on the HV Rectifier board and insert it in J103 for 50 Hz operation or J102 for 60 Hz operation.

Locate P105 on the HV Rectifier board and insert it in J105 for 50 Hz operation or J106 for 60 Hz operation.

Remove the shorting straps from the HV Rectifier board. Reconnect the power supply fan connector and replace the HV front panel.

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni		Omni		

9. CONSOLE

Position the console table against the wall in the operator's room. Position the "Operator equipment cabinet" underneath the table. The table height can be adjusted.

Figure 11 - Console

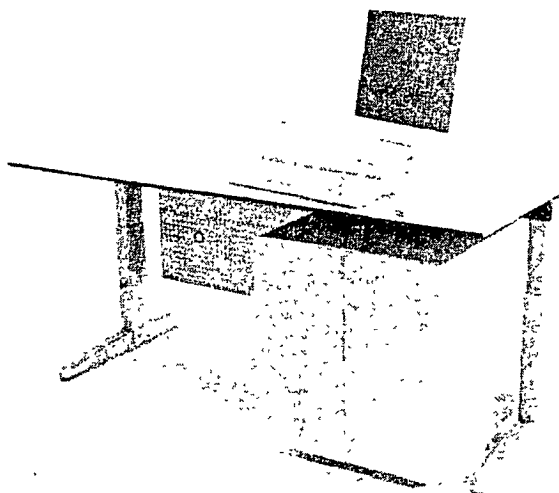
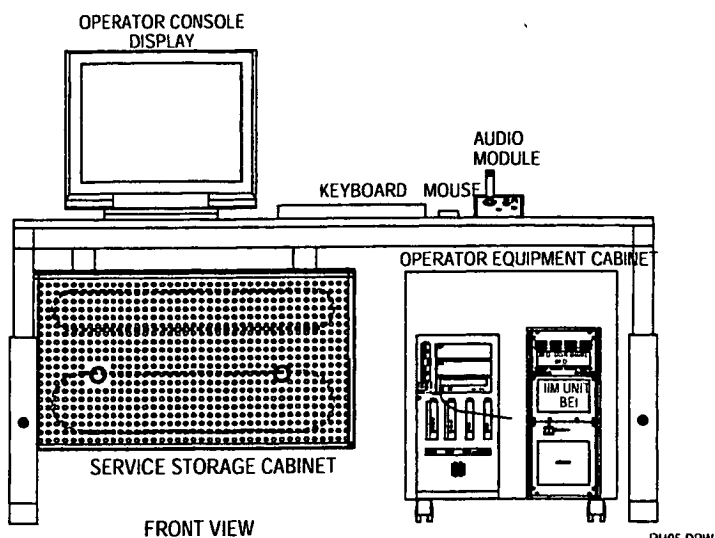


Figure 12 - Console with units



Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

Section 3

Cabinet installation

Contents

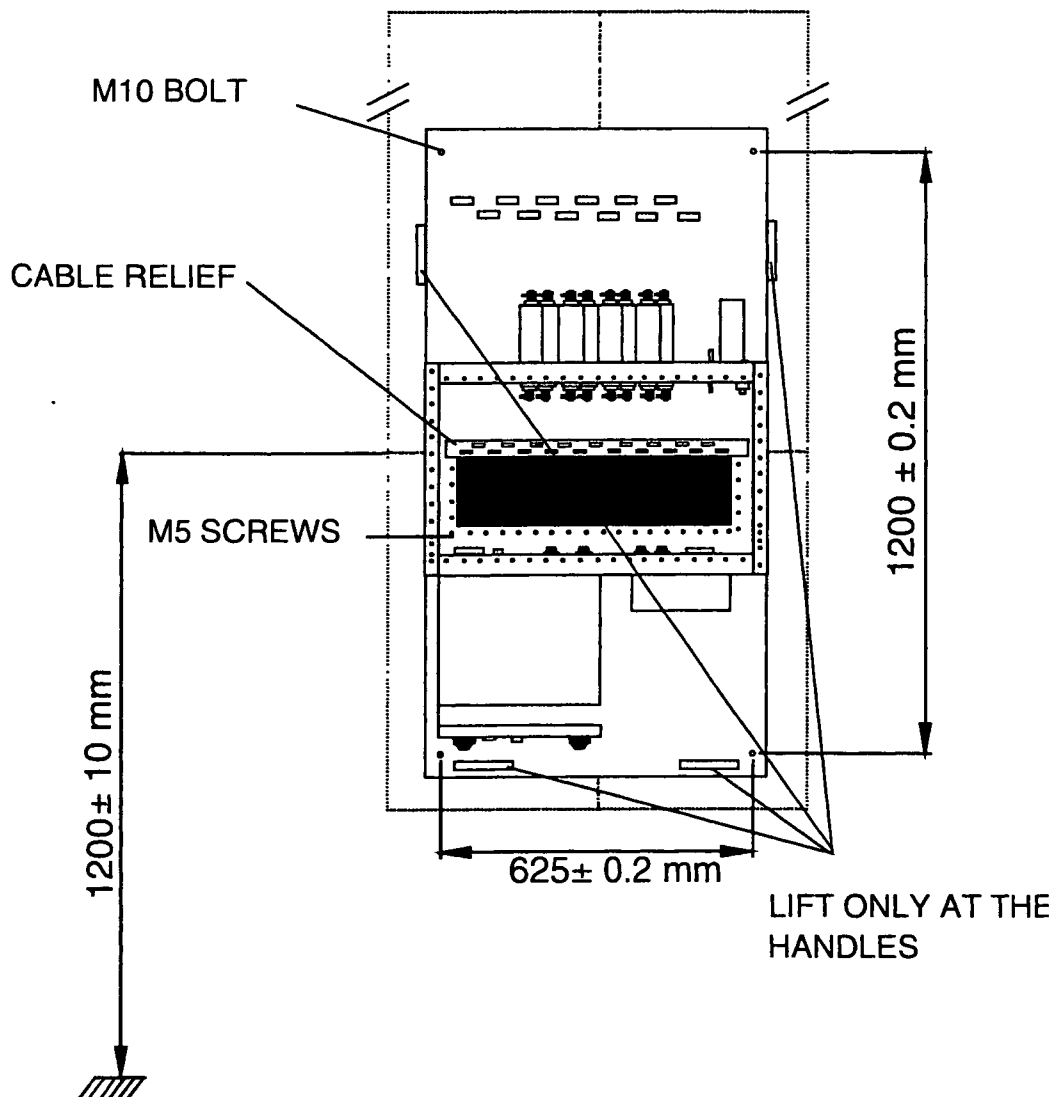
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Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

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2. Remove the upper aluminum covers from the System Filter Box to gain access to the handles. Also remove the middle copper cover.
3. Clean the contact area's of the RF cage interface and of the system filter box.
4. Hang the System Filter Box on the M10 bolts (do not tighten) prepared by the RF supplier. Only lift at the handles.
5. Mount all M5 screws with washers delivered by RF cage supplier.
6. Tighten the M10 nuts.

Figure 1 - Mounting the System Filter Box

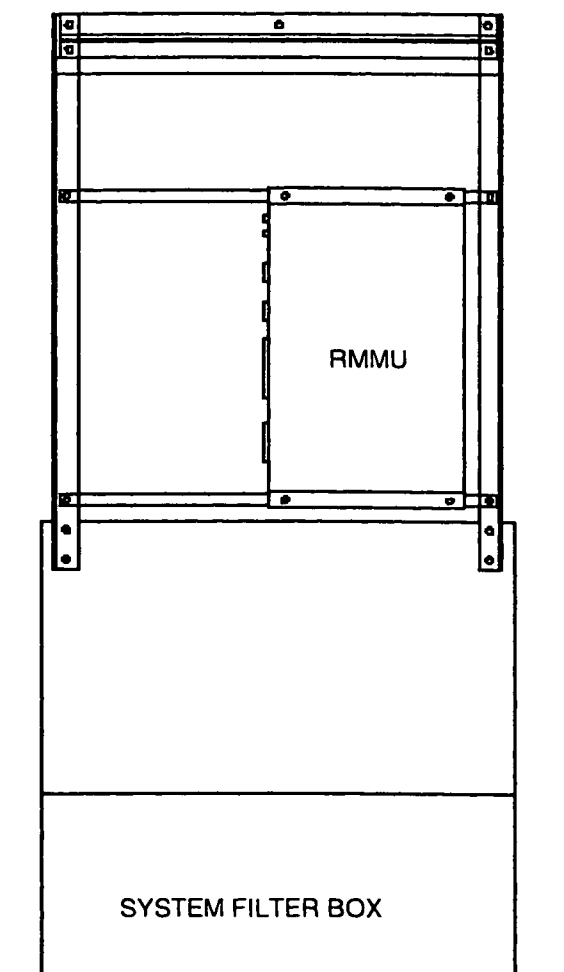


Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

2. MOUNTING THE REMOTE MAGNET MONITORING UNIT (RMMU)

The RMMU must be mounted on the topframe of the System Filter Box (SFB).

Figure 2: Position of RMMU



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3.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

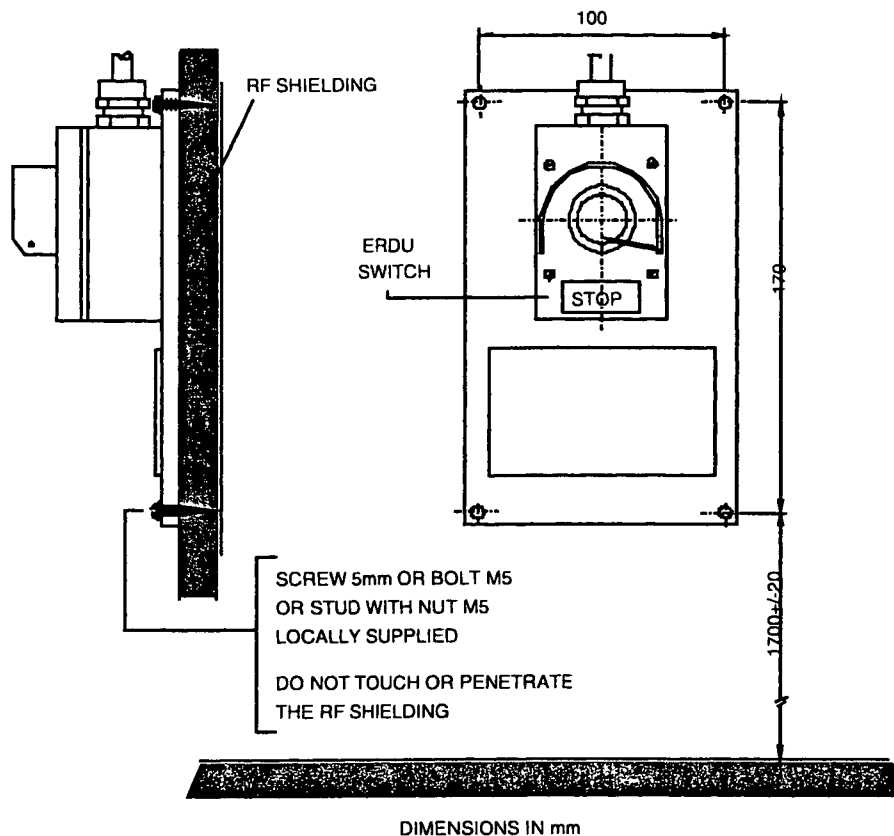
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One of the ERDU switches must be mounted inside the RF shielded room and one in the control room. Mount the ERDU switch inside the RF room to the planned position (see drawings of the planning department). Ask the RF room supplier how to mount the ERDU switch in order not to damage the RF shield

CAUTION

Be sure not to penetrate or make contact with the RF shielding when installing the ERDU switch inside the RF room.

Figure 3: Mounting the ERDU switch

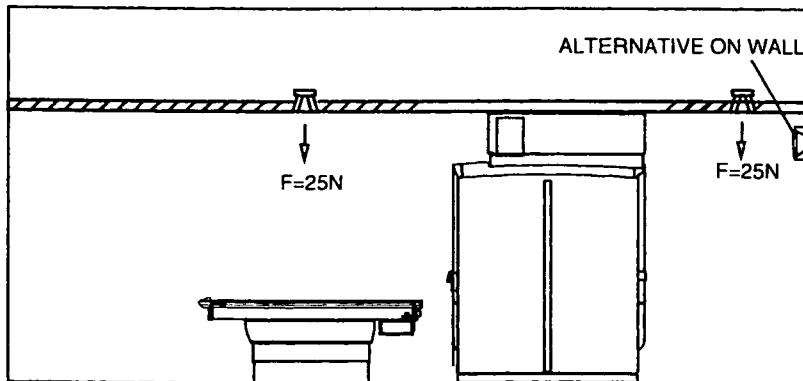


Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

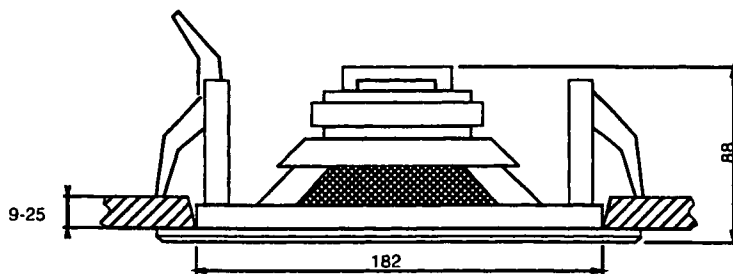
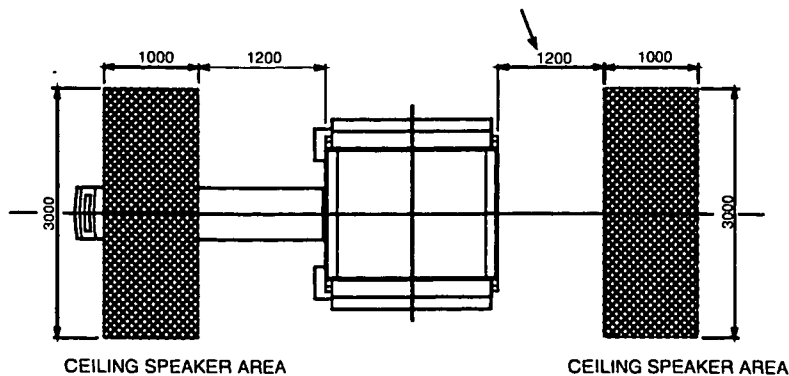
4. MOUNTING THE CEILING SPEAKERS

Two ceiling speakers have to be mounted. One above the patient support and the other one at the rear of the magnet (see figure). The distance of 1.20m is an average in order to obtain a good sound quality on both sides of the magnet. The minimum distance at the rear side is 0.3m. In that case, the quality is decreases but is still acceptable. As an alternative, it is possible to mount the speaker on the wall. Use one of the delivered build-on boxes.

Figure 4 - Mounting positions of the ceiling speakers



MINIMAL 550mm IN CASE OF MINIMAL ROOM LAYOUT



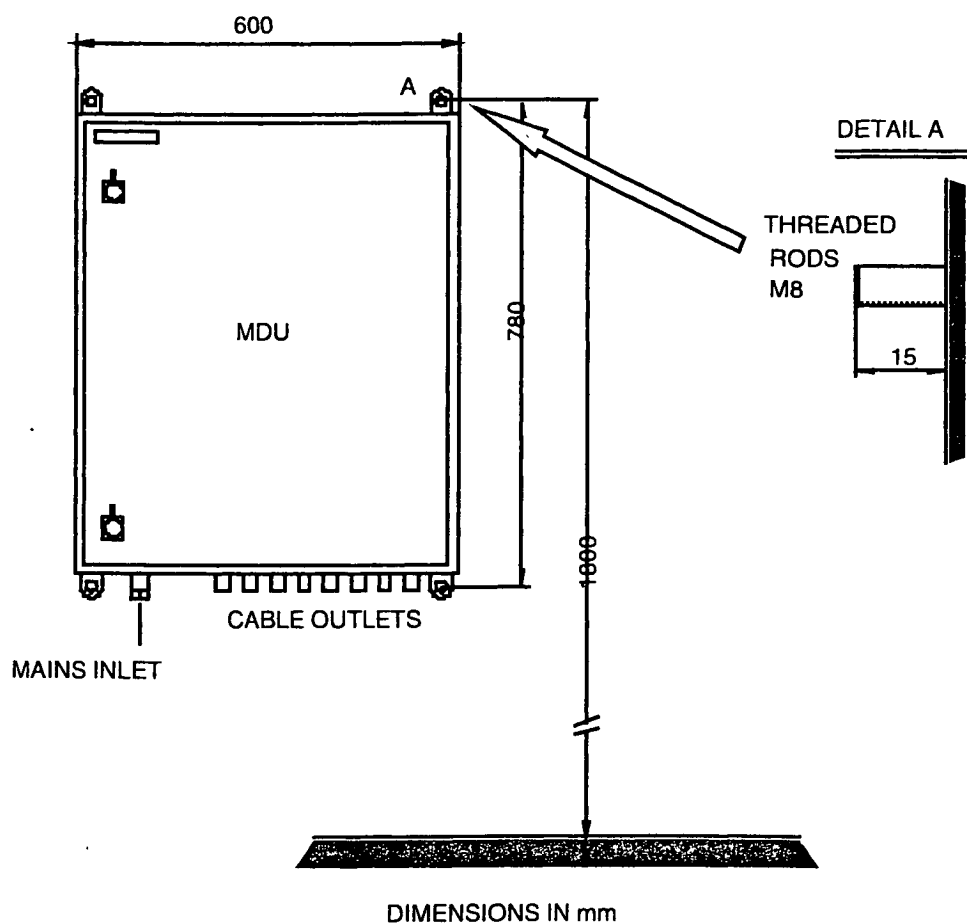
Dimensions in mm

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

5. MOUNTING THE MAINS DISTRIBUTION UNIT (MDU)

The MDU must be mounted to four threaded rods mounted in the technical room. Arrange a local electrician to connect the hospital mains to the MDU

Figure 5 - Mounting the MDU

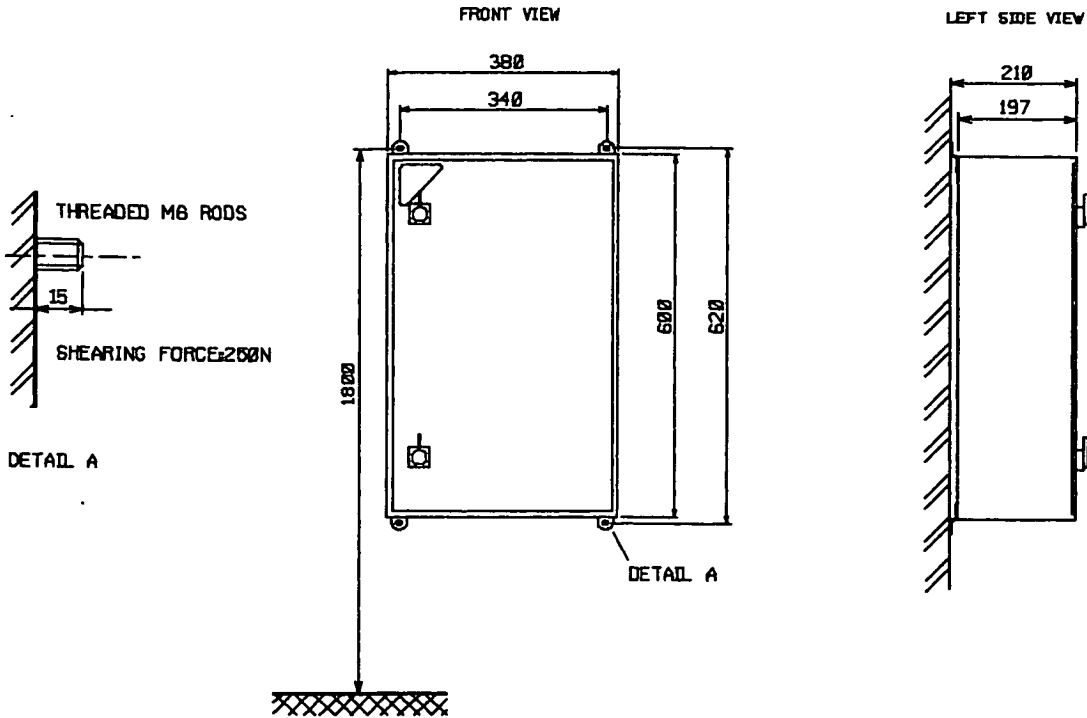


Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

6. MOUNTING THE MAINS DISTRIBUTION EXTENSION UNIT (MDE)

Also the MDU extension must be mounted to four studs mounted in the technical room. Arrange an electrician to connect the hospital mains to the MDU.

Figure 6: MDU extension

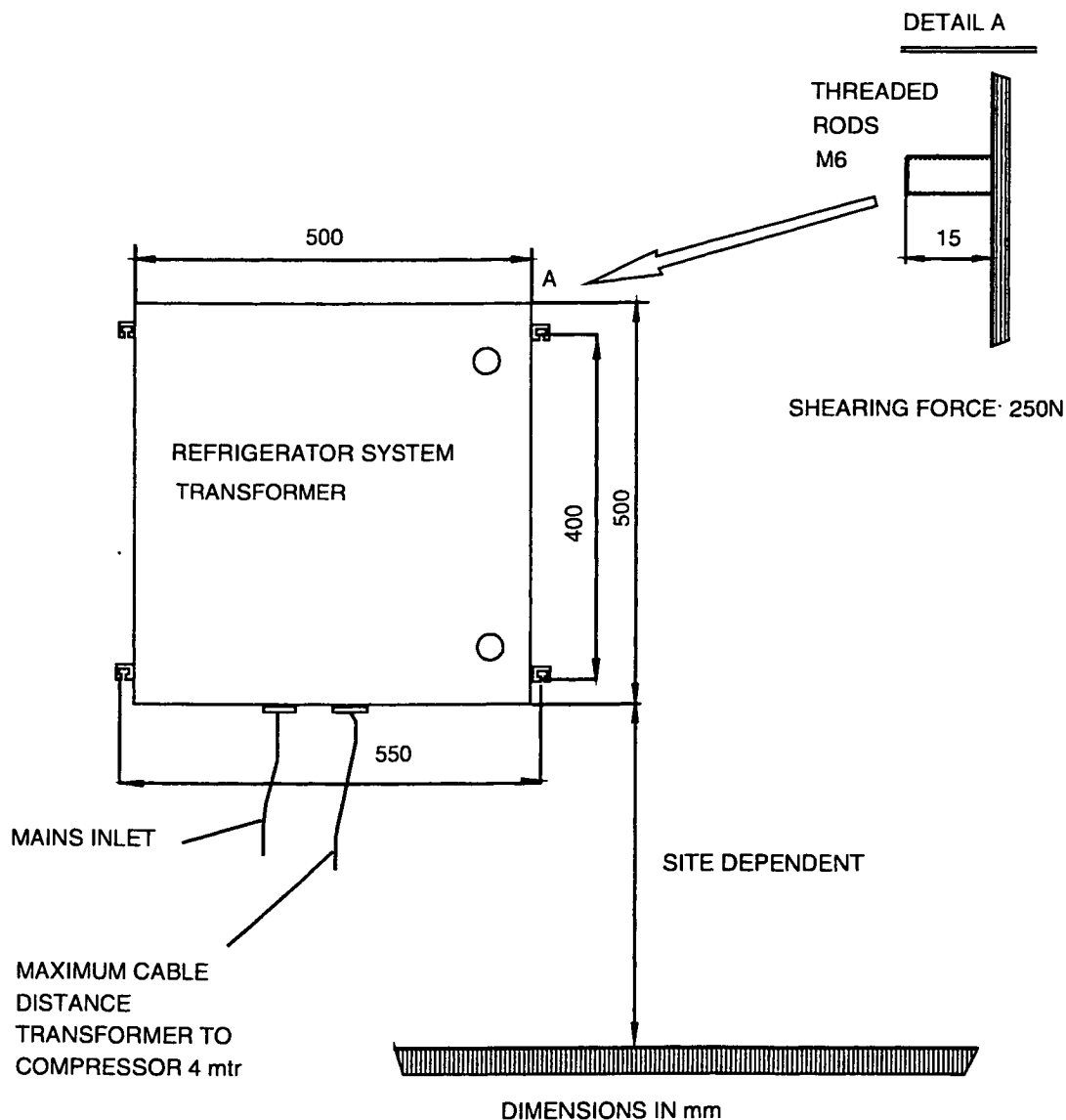


Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

7. MOUNTING THE REFRIGERATOR SYSTEM TRANSFORMER (NOT FOR USA)

The refrigerator system transformer must be mounted to four threaded rods in the technical room.

Figure 7: Mounting the refrigerator system transformer



FH127 DRW

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

8. POSITIONING CABINETS

Position all cabinets into the technical room according to the Site-Planning drawings. For dimensions of cabinets and service area consult the Intera Planning Reference Book.
Level all cabinets using a spirit level and tighten all feet.

8.1. GRADIENT AMPLIFIER 274

Unpacking:

WARNING

Under no circumstances move the packing if it is partly unpacked. When the front riser assembly is removed, the front wheels will rest on metal plates. The rear wheels are elevated and the cabinet will rest on the riser blocs. The cabinet then will slightly tilted forward. To move the cabinet with its packing in this configuration is very dangerous.

The riser is marked with:

**DO NOT DISCARD
DO NOT TRANSPORT UNLESS THIS ASSEMBLY IS INSTALLED**

1. Remove all screws circled in red. Remove front panel and the right side panel of the crate.
2. Remove the front riser assembly. This supports the weight of the cabinet and protects the casters during shipment.
3. Remove the ramp assembly and its support, attached at the inner right side panel. Place the heavy wooden bar halfway under the ramp for additional support. Attach the ramp assembly into the slots at the front of the pallet.
4. Use the nylon straps to pull the amplifier down the ramp to floor level.

CAUTION

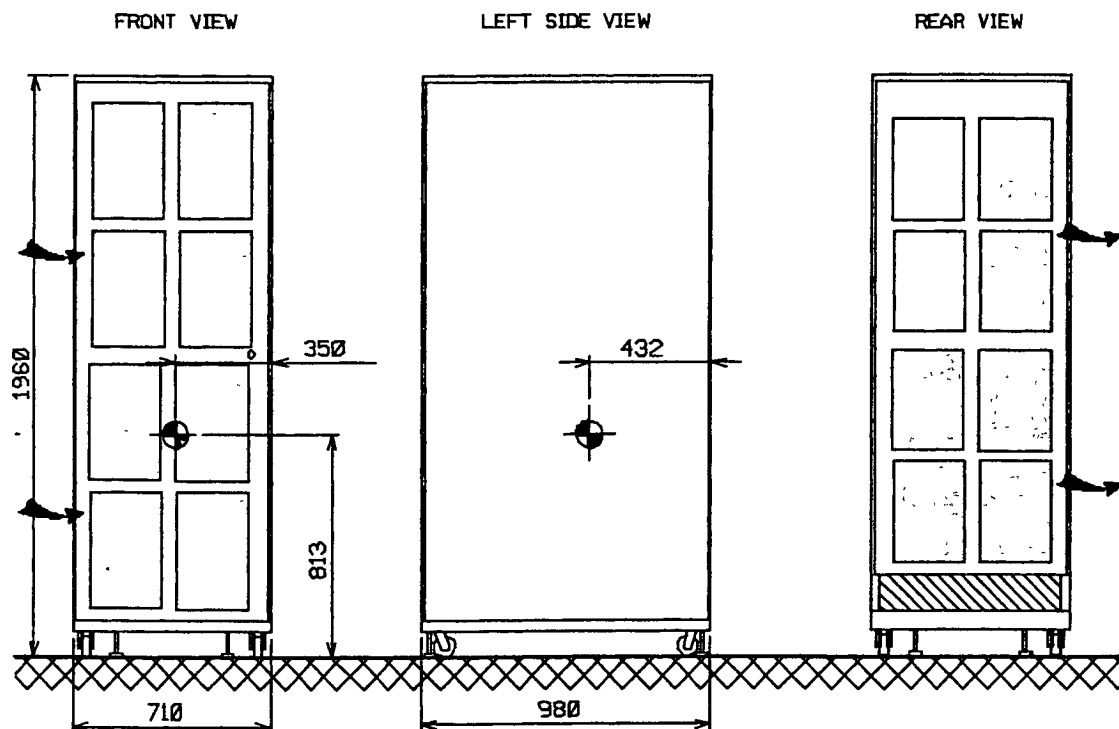
The gradient amplifier 274 may only be transported in a vertical position. The unit weight is 550 Kg. If it is needed to transport the unit horizontally, than first remove the 12 power modules. Otherwise you will damage the back plane of the cabinet. For the location of these power modules refer to drawing Z3-6.2.1 of the drawing manual.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

8.1.1. INSTALLATION FOR THE POWER GRADIENTS

1. Move the cabinet to where it will be installed. The cabinet including the swivel wheels is 196 cm high.
2. Remove the straps and plastic covering from the cabinet.
3. Adjust the cabinet levelers for stability.

Figure 8: Copley 274 gradient amplifier



Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

8.2. INSTALLATION OF THE MASTER GRADIENTS

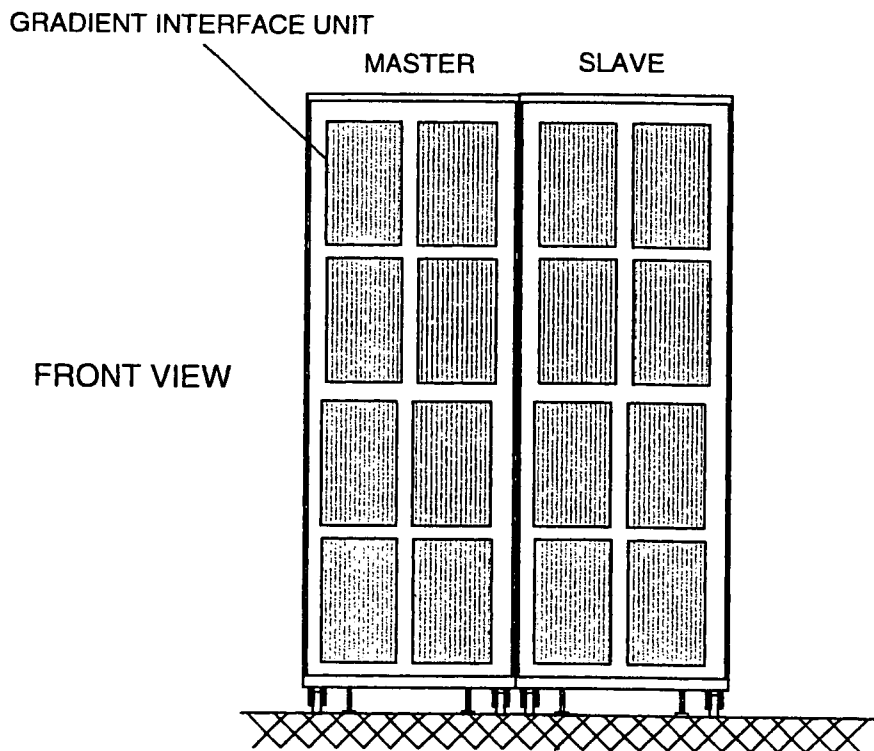
In this case there are two 274-gradient amplifiers a Master and a Slave model. The Master 274 amplifier is recognizable by the already installed gradient interface rack inside the cabinet in the upper position. The Slave version does not have this rack.

The Dual mode interface board DIR is supplied in a separate box. This board must be installed in the Master 274 amplifier. The existing internal cables must be shifted.

Move the cabinets to where it will be installed. **Master cabinet must be installed left and the Slave right faced from the front side.** The cabinets including the swivel wheels are 196 cm high. **The master cabinet and slave cabinet must be bolted together for stability and in order to guarantee EMC.**

Remove the straps and plastic covering from the cabinets.

Figure 9: Master Slave front view

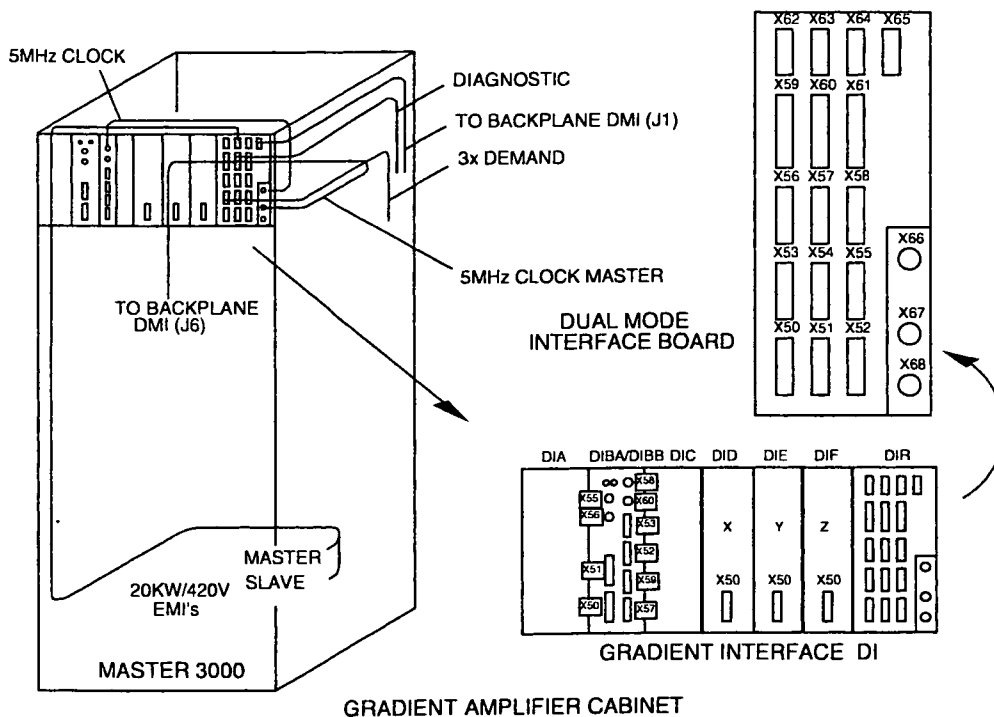


Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

Procedure master cabinet

1. Remove the 4 screws of the right side panel of the Master gradient cabinet. Remove this side panel and also the four side panel brackets and the suspension bracket inside the cabinet.
2. **Do not remove the EMC tape.**
3. Disconnect the three control cables on the WFG/ECC boards DI.X50.
4. Check the jumper settings of the Controller (DIB.)- and WFG/ECC DFEC3 interface boards, according Section 17 paragraph "PRE POWER-UP CHECKS PROGRAMMING"

Figure 10 Power gradients cabinet after rebuilding.

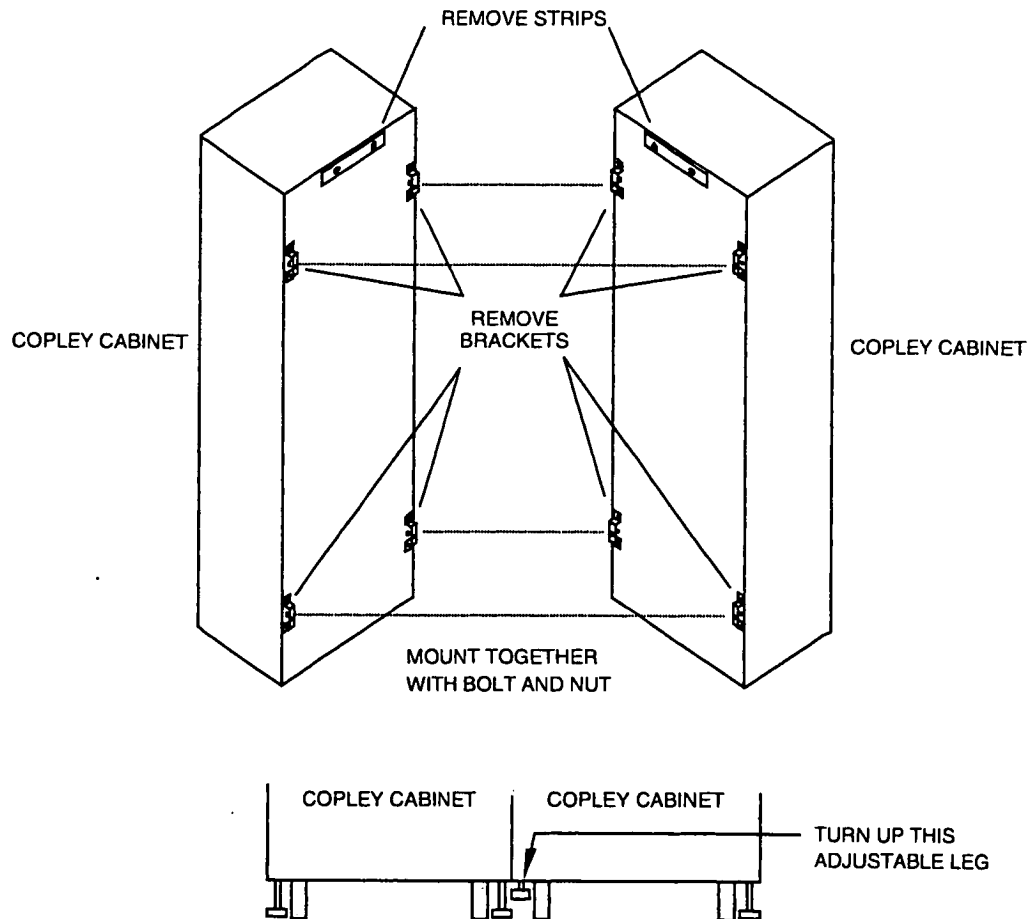


5. Unscrew the two right side blind plates from the gradient interface rack (EC). Check jumper settings on the dual mode interface board (DIR) jumpers W1, W2, W4, W6 in position 1-2 and jumpers W3, W5, W7 in position 2-3. Install DIR board on the right side.
6. Now shift and re-label the following internal cables:
 - DIBX58 to DIRX67 5 MHz clock
 - DIBX51 to DIRX60 Diagnostic cable
 - DIBX52 to DIRX63 EMI 20kW/420V master programming cable
7. Install the new supplied cables:
 - DIDX50 - DIRX50 Z demand input cable
 - DIEX50 - DIRX51 Y demand input cable
 - DIFX50 - DIRX52 X demand input cable
 - DIBX52 - DIRX62 EMI 20kW/420V programming cable
 - DIRX59 - DIBX51 Diagnostic cable
 - DIRX66 - DIBX58 BNC clock cable
 - DIRX65 - DM1J1 Power supply cable

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

8. Reconnect the existing master cabinet cables and re-label them:
 X demand cable (old label DIFX50-X) to DIRX55.
 Y demand cable (old label DIEX50-Y) to DIRX54.
 Z demand cable (old label DIDX50-Z) to DIRX53.
9. Bolt both cabinets together on the 4 released tags of the side-panel brackets. Use 4 x M 5 screws/nuts and two knurled washers on each side of the cabinets-tags. The new assembly is also EMC proofed.

Figure 11: Mount together Master and Slave

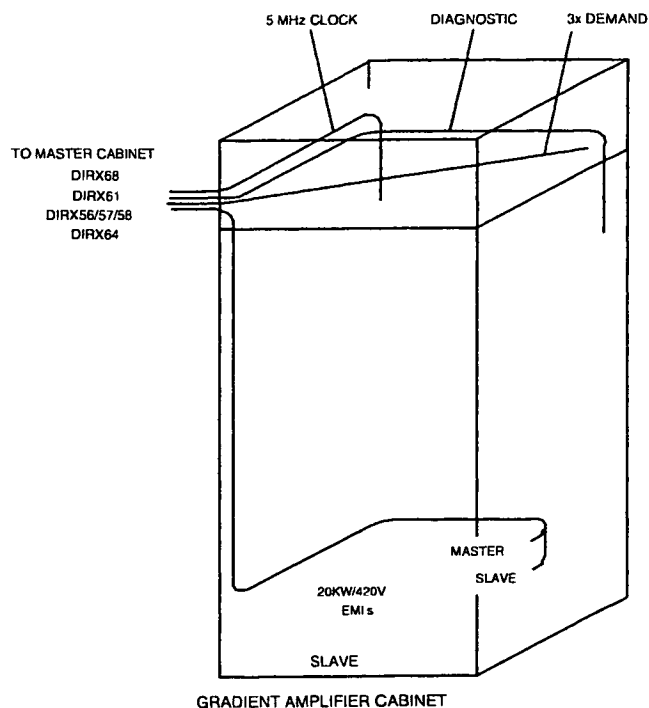


10. Adjust the cabinets levelers for stability. Turn up one adjustable leg.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

Procedure slave cabinet:

1. Disconnect the tie-wrap on the right side of the Slave gradient cabinet, which fixes the demand cables and shift the three demand to the left side of the cabinet.

Figure 12: Installation second cabinet.

HH89 WPG

2. Connect the slave cabinet cables to the Dual mode interface board and re-label them:
 - X demand cable (old label DIFX50-X) to DIRX58.
 - Y demand cable (old label DIFX50-Y) to DIRX57.
 - Z demand cable (old label DIFX50-Z) to DIRX56.
 - 5 MHz clock cable (old label DIBX58) to DIRX68.
 - Diagnostic cable (old label DIBX51) to DIRX61.
 - EMI 20kW/420V programming cable (old label DIBX52) to DIRX64.
- Do not forget to install the earth cable DX0 (master) to DX0 (slave) as described in section 5.

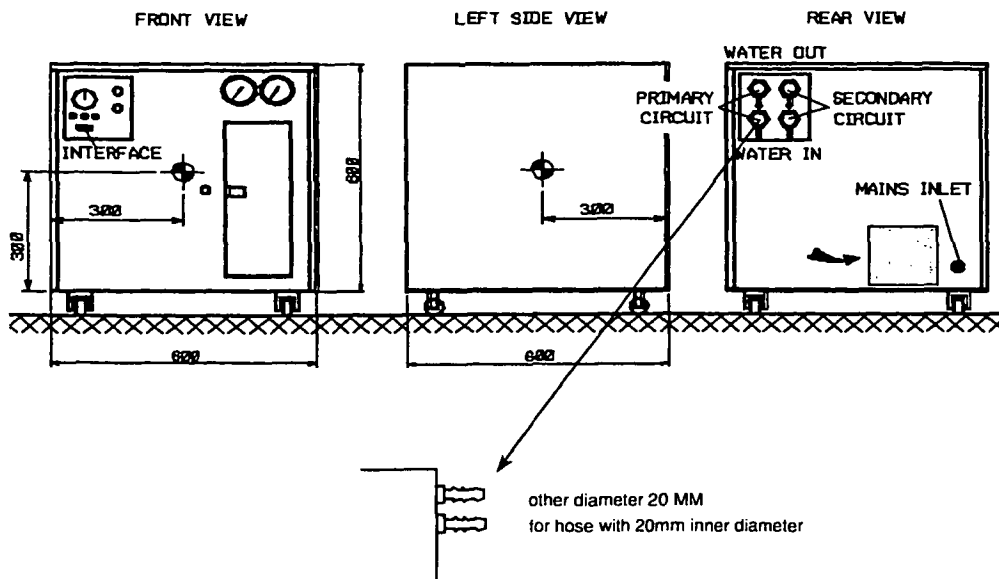
Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

8.3. HEAT EXCHANGER (ONLY POWER AND MASTER GRADIENTS)

Connect the hospital cooling water to the primary in and out fittings.

Connect the cooling hoses from the gradient coil to the secondary in and out fittings.

Figure 13 - Head exchanger

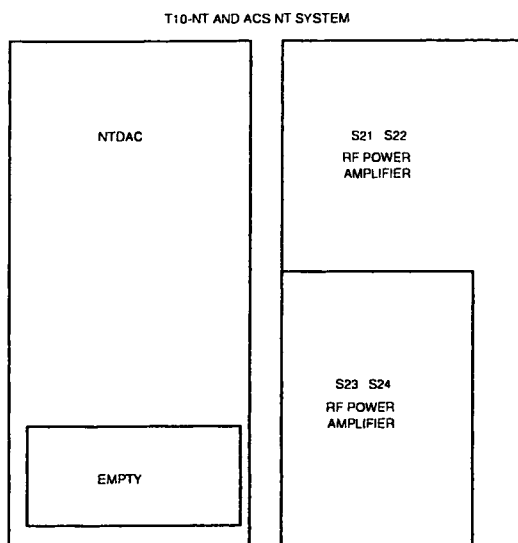


Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

8.4. RF-AMPLIFIER

For Intera 1.0T and 1.5T the RF amplifier must be positioned next to the NTDAC.

Figure 14: Location RF Amplifier



GR01 WPG

Release independent, a new (smaller) RF amplifier will be supplied.
This is the S23 (Intera 1.5T) and the S24 (Intera 1.0T).

NOTE

*The Mains Power (three phase) for the S21, S22 and MR5002 RF amplifiers comes from the NT-DAC.
However, the Mains Power (single phase) for the S23 and S24 RF amplifiers comes directly from the MDU.*

NOTE

The RF-Output on the S23 and S24 RF amplifiers is located on the rear of the amplifier.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

9. LINE FREQUENCY (50/60 HZ) SELECTION (S21 AND S22 AMPLIFIERS ONLY).

The S21-64 and S22-42 RF amplifiers must be adjusted to 50 or 60 Hz..

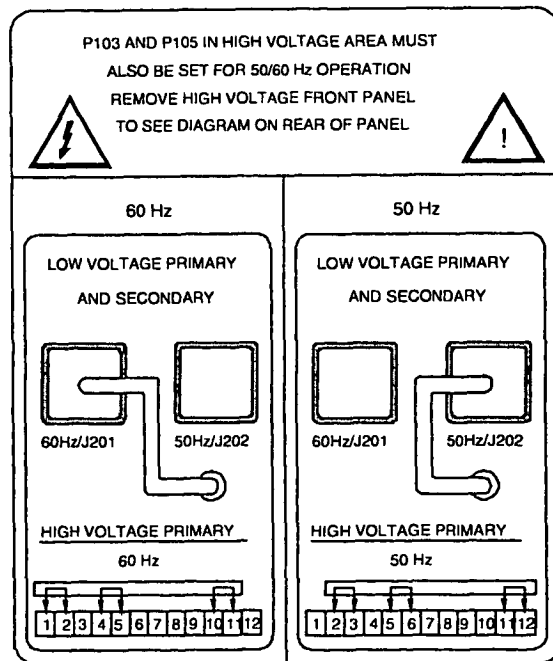
CAUTION

S21-64 and S22-42 RF Amplifier Line Frequency (50/60 Hz) Selection.

The S21/22 RF amplifier incorporates ferro-resonant supplies, which must be configured for either 50 Hz or 60 Hz operation. To re-configure the amplifier from one frequency to the other:

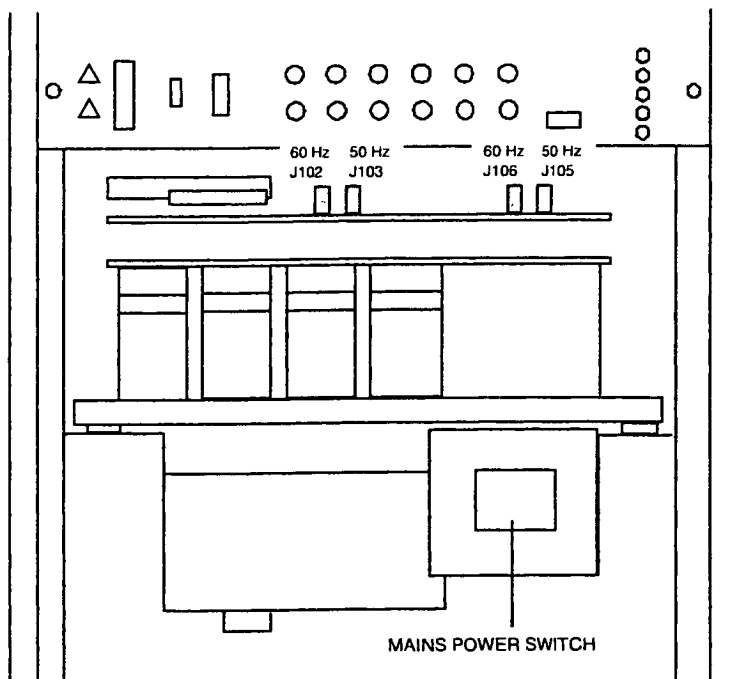
1. If not yet done, switch off the mains power to the RF amplifier. Disconnect the mains power plug of the RF amplifier from the NT-DAC.
2. Remove the 50/60 Hz service panel from the rear of the RF amplifier and follow the directions located on the panel. Replace the panel when complete.

Figure 15: Rear Panel screening



3. Remove the panel covering the HV assembly. This is the lower front panel. Take care to unplug the power supply fan when removing the panel. Its connector is located behind the panel on the lower right hand side. Attach a shorting strap between PA HV+ and PA HV- on the HV Rectifier board. Also connect a shorting strap between IPA HV+ and IPA HV- on the HV Rectifier board.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

Figure 16: HV Assembly

Locate P103 on the HV Rectifier board and insert it in J103 for 50 Hz operation or J102 for 60 Hz operation.

Locate P105 on the HV Rectifier board and insert it in J105 for 50 Hz operation or J106 for 60 Hz operation.

Remove the shorting straps from the HV Rectifier board. Reconnect the power supply fan connector and replace the HV front panel.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

10. CONSOLE

Position the console table against the wall in the operator's room. Position the "Operator equipment cabinet" underneath the table. The table height can be adjusted.

Figure 17 - Console

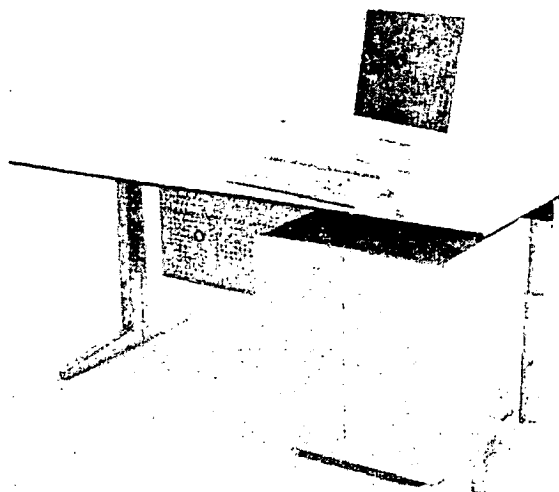


Figure 18 – Console with units

